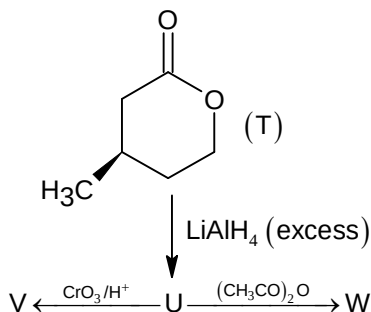


SECTION – A

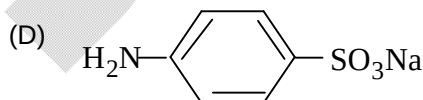
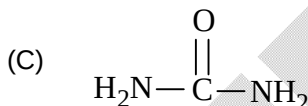
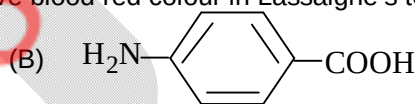
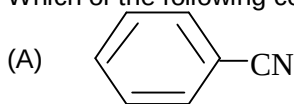
(One or More than one correct type)

This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

22. With reference to the scheme given, which of the following statement(s) about T, U, V and W is/are correct?



- (A) T is soluble in hot aqueous NaOH
 (B) U is optically active
 (C) Molecular formula of W is $\text{C}_{10}\text{H}_{18}\text{O}_4$
 (D) V gives effervescence on treatment with aqueous NaHCO_3
23. Which of the following compound(s) does not give blood red colour in Lassaigne's test ?



24. In the given question K_a values of five weak acids are given. K_a for $\text{HA} = 10^{-8}$, $\text{HB} = 10^{-6}$, $\text{HC} = 2 \times 10^{-8}$, $\text{HD} = 10^{-10}$, $\text{HE} = 10^{-7}$.
 Select the correct statement(s) as per the information. Given : $\log 2 = 0.3$.
 (A) Among NaA, NaB, NaC, NaD and NaE, the NaD is most extensively hydrolysed.
 (B) Among NaA, NaB, NaC, NaD and NaE, the NaB is most extensively hydrolysed.
 (C) pH of 0.1 (M) aqueous solution of NaB is approximately 9.5.
 (D) 0.1 (M) solution of HE and 0.01 (M) solution of HB are isohydric to each other approximately.

SECTION – B

(Numerical Answer Type)

This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.

25. How many of the following reagents reduce nitrobenzene in azobenzene?
 LiAlH_4 , Sn / HCl , $\text{Zn} + \text{NaOH} / \text{MeOH}$, NaBH_4 , NH_4HS
26. Difference between n^{th} and $(n + 1)^{\text{th}}$ Bohr's radius of H-atom is equal to its $(n - 1)^{\text{th}}$ Bohr's radius. The value of n is